

PRACTICAL FILE

PYTHON SEM-1

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B.Sc.(Hons) CS

Q1- Write a program to find the roots of a quadratic equation.

Program:

```
from math import sqrt
print("To find root of quadratic equation enter a,b,c")
while True:
    try:
        a=eval(input('Enter a:'))
        if type(a) not in [int,float] or a==0 :
            print('Entered data is not correct acc. to a')
        else:
            while True:
                b=eval(input('Enter b:'))
                c=eval(input('Enter c:'))
                if type(b) in [int,float] and type(c) in [int,float]:
                    break
                else:
                    print('Enter a integer')
            break
    except:
        print('Enter Numbers only')
D=b**2-(4*a*c)
if D>=0:
    print('so roots of Equation:',(-b+sqrt(D))/(2*a),end=',')
    print((-b-sqrt(D))/(2*a))
else:
    print('as D<0 so no real roots')
```

Output:

```
===== RESTART: C:\Users\Shruti\Desktop\Niket Aggarwa
To find root of quadratic equation enter a,b,c
Enter a:0
Entered data is not correct acc. to a
Enter a:xyz
Enter Numbers only
Enter a:-1
Enter b:2
Enter c:5
so roots of Equation: -1.4494897427831779,3.449489742783178
```

```
===== RESTART: C:\Users\Shruti\Desktop
To find root of quadratic equation enter a,b,c
Enter a:5
Enter b:20
Enter c:20
so roots of Equation: -2.0,-2.0
```

```
===== RESTART: C:\Users\Shruti\Desktop\Ni
To find root of quadratic equation enter a,b,c
Enter a:1
Enter b:2
Enter c:3
as  $D < 0$  so no real roots
|
```

Q2- Write a program to accept a number 'n' and

- a. Check if 'n' is prime.
- b. Generate all prime number till 'n'.
- c. Generate first 'n' prime number.

This program may be done by using functions.

Program:

```
def prime(n):
    for i in range(2,n):
        if n%i==0:
            return False
    return True
while True:
    try:
        n=int(input('Enter a number n so to perform some prime number task:'))
        if n<=0:
            print('Enter interger greater than 0')
        else:
            break
    except:
        print('Enter interger greater than 0')
if prime(n):
    print('a)',n,'is prime')
else:
    print('a)',n,'is not prime')
print('b) All prime number till',n,'is:',end='')
for i in range(2,n+1):
    if prime(i):
        print(i,end=' ')
print()
print('c) first',n,'prime number are:',end='')
c=0;a=2
while c!=n:
    if prime(a):
        print(a,end=' ')
        c+=1
    a+=1
```

Output:

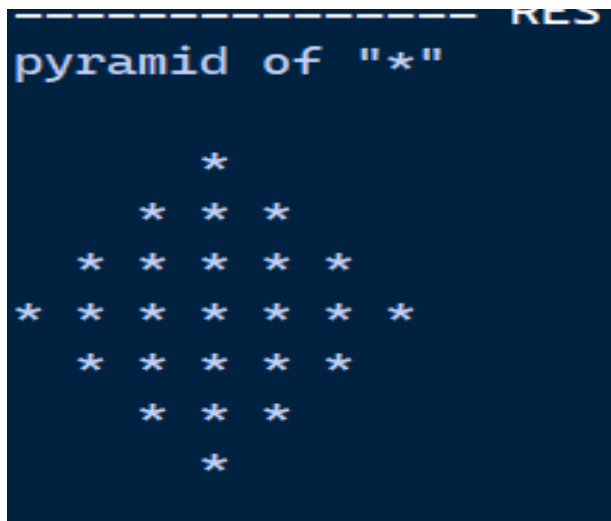
```
===== RESTART: C:\Users\Shruti\Desktop\Niket A
Enter n to all purpose:g
Enter interger greater than 0
Enter n to all purpose:-1
Enter interger greater than 0
Enter n to all purpose:2.5
Enter interger greater than 0
Enter n to all purpose:10
a) 10 is not prime
b) All prime number till 10 is:2 3 5 7
c) first 10 prime number are:2 3 5 7 11 13 17 19 23 29
```

Q3-Write a program to create a pyramid of the character '*' and a reverse pyramid.

Program:

```
print('pyramid of "*"\\n')
n=2
for i in range(3):
    print(" "*(6-i*2),end="")
    for j in range(1,n):
        print("*",end=' ')
    n=n+2
    print()
for i in range(3,8):
    print(" "*(i*2-6),end="")
    for j in range(1,n):
        print('*',end=' ')
    n=n-2
    print()
```

Output:



Q4- Write a program that accepts a character and performs the following:

- print whether the character is a letter or numeric digit or a special character.
- if the character is a letter, print whether the letter is uppercase or lowercase.
- if the character is a numeric digit, prints its name in text(e.g. if input is 9, output is Nine)

Program:

```
print("To check the type and details of entered character")
d={1:'One',2:'Two',3:'Three',4:'Four',5:'Five',6:'Six',7:'Seven',8:'Eight',9:'Nine',0:'Zero'}
a=input('Enter:')
if a.isdigit():
    print(a,'is numeric digit')
    if len(a)==1:
        print(a,'is',d[int(a)])
elif a.isalpha():
    print(a,'is a letter')
    if a.isupper():
        print(a,'is in uppercase')
    elif a.islower():
        print(a,'is in lowercase')
    else:
        print("Mix of both lower and upper")
elif a.isalnum()!=True and a.isspace()!=True:
    print(a,'is a special character')
else:
    print('Not recognised')
```

Output:

```
To check the type and details of entered character
Enter:niket
niket is a letter
niket is in lowercase

===== RESTART: C:\Users\Shruti\Desktop\Nik
To check the type and details of entered character
Enter:NIKET
NIKET is a letter
NIKET is in uppercase

===== RESTART: C:\Users\Shruti\Desktop\Nik
To check the type and details of entered character
Enter:2334
2334 is numeric digit

===== RESTART: C:\Users\Shruti\Desktop\Nik
To check the type and details of entered character
Enter:2
2 is numeric digit
2 is Two

===== RESTART: C:\Users\Shruti\Desktop\Nik
To check the type and details of entered character
Enter:%$
%$ is a special character
```

Q5- Write a program to perform the following operations on a string

- a. Find the frequency of a character in a string.
- b. Replace a character by another character in a string.
- c. Remove the first occurrence of a character from a string.
- d. remove all occurrences of a character from string.

Program:

```
print('To perform some string operation enter string and a character from that string')
a=input('Enter a String:')
c=input('Enter character:')
print('a) Frequency of',c,'in',a,'is:',a.count(c),'\n')
b=input('b) By which character you want to replace '+c+':')
print('New string is;',a.replace(c,b),'\n')
print('c) Removing first occurrence')
print('Original string',a)
x=a.find(c)
if x==-1:
    print('Character is not in string')
    print('No change\n')
else:
    print('New string:',a[:x]+a[x+len(c):],'\n')
print('d) Removing all occurrence')
y=0;d=a
for i in range(a.count(c)):
    x=a.find(c)
    a=a[:x]+a[x+len(c):]
    y+=1
if y==0:
    print('Same is old string:',a)
else:
    print('Old string:',d)
    print('New string:',a)
```

Output:

```
===== RESTART: C:\Users\Shruti\Desktop\Niket Aggarwal\bsc hons cs\pra
To perform some string opeartion enter string and a character from that string
Enter a String:xyz123okokBYEbye
Enter character:o
a) Frequency of o in xyz123okokBYEbye is: 2

b) By which character you want to replace o:ok
New string is; xyz123okkokkBYEbye

c)Removing first occurence
Original string xyz123okokBYEbye
New string: xyz123kokBYEbye

d)Removing all occurrence
Old string: xyz123okokBYEbye
New string: xyz123kkBYEbye
```

```
===== RESTART: C:\Users\Shruti\Desktop\Niket Aggarwal\bsc hons cs\pra
To perform some string opeartion enter string and a character from that string
Enter a String:xyz123okokBYEbye
Enter character:A
a) Frequency of A in xyz123okokBYEbye is: 0

b) By which character you want to replace A:B
New string is; xyz123okokBYEbye

c)Removing first occurence
Original string xyz123okokBYEbye
Character is not in string
No change

d)Removing all occurrence
Same is old string: xyz123okokBYEbye
```

Q6- Write a program to swap the first n character of two string.

Program:

```
print('Swaping first n character from two string')
a=input('Enter string1:')
b=input('Enter string2:')
while True:
    try:
        n=int(input('Enter n character to be swap:'))
        if n<0:
            print('Enter positive number')
        else:
            break
    except:
        print('Enter number only')
x=b[:n]+a[n:]
b=a[:n]+b[n:]
a=x
print('NEW after swaping')
print('String1:',a)
print('String2:',b)
```

Output:

```
===== RESTART: C:\Users\Shruti\De
Swaping first n character from two string
Enter string1:sjfbjfbe
Enter string2:ebfjefjewfbew
Enter n character to be swap:-1
Enter positive number
Enter n character to be swap:0
NEW after swaping
String1: sjfbjfbe
String2: ebfjefjewfbew

===== RESTART: C:\Users\Shruti\De
Swaping first n character from two string
Enter string1:fbdjfewjfwef
Enter string2:fbjwbjfwdwd
Enter n character to be swap:5
NEW after swaping
String1: fbjwbewjfwef
String2: fbdjffjwdwd
```

Q7- Write a program that accepts two strings and returns the indices of all the occurrences of the second string in the first string as a list. If the string is not present in the first string then it should return -1.

Program:

```
print('Enter two string to check that the second string occur in first or not')
def checkstr(a,b):
    l=[]
    if a.count(b)==0:
        return -1
    else:
        c=len(b)
        for i in range(len(a)):
            if a[i:i+c].lower()==b.lower():
                l.append(i)
        return l
a=input('Enter first string:')
b=input('Enter second string:')
l=checkstr(a,b)
if l==[-1]:
    print('No occurrence of',b,'in',a,'(-1)')
else:
    print(b,'occurs in',a,'on indices:',l)
```

Output:

```
Enter two string to check that the second string occur in first or not
Enter first string:niketagggarwal
Enter second string:a
a occurs in niketagggarwal on indices: [5, 8, 11]

===== RESTART: C:\Users\Shruti\Desktop\Niket Aggarwal\bsc hons c
Enter two string to check that the second string occur in first or not
Enter first string:niket
Enter second string:b
No occurrence of b in niket (-1)
```

Q8- Write a program to create a list of cubes of only the even integers appearing in the input list(may have elements of other types also) using the following:

a)'for' loop.

Program:

```
while True:
    l=eval(input('Enter a list:'))
    if type(l)==list:
        break
    else:
        print('Enter list only\n')
cl=[]
for i in l:
    if type(i)==int and i%2==0:
        a=i**3
        cl.append(a)
print('Cube of even number in given list:',cl)
```

b) list comprehension.

Program:

```
while True:
    l=eval(input('Enter a list:'))
    if type(l)==list:
        break
    else:
        print('Enter list only\n')
ccl=[a**3 for a in l if type(a)==int and i%2==0]
print('Cube of even number in given list:',ccl)
```

output

```
===== RESTART: C:\Users\Shruti\Desktop
Enter a list:122
Enter list only

Enter a list:[12,4,3,'fe',(1,2),5]
Cube of even number in given list: [1728, 64]

===== RESTART: C:\Users\Shruti\Desktop
Enter a list:[1,2,3,4,5]
Cube of even number in given list: [8, 64]
```

Q9- Write a program to read a file and

- Print the total number of characters, words and lines in the file.
- Calculate the frequency of each character in the file. Use a variable of dictionary type to maintain the count.
- Print the words in reverse order.
- Copy even lines of the file to a file named 'File1' and odd lines to another file named 'File2'.

Program:

```
print('File Name:Text.txt')
```

```
try:
```

```
    f=open("Text.txt",'r')
```

```
    f.close()
```

```
except:
```

```
    print("\nFile doesnot exist first create it')
```

```
    n=eval(input('How many lines do you want to add:'))
```

```
    f=open("Text.txt",'a')
```

```
    for i in range(1,n+1):
```

```
        a=input('Enter line'+str(i)+':')
```

```
        f.write(a+"\n')
```

```
    print('File created\n')
```

```
    f.close()
```

```
#a
```

```
print("a)Total no. of character,words and lines in the file")
```

```
f=open("Text.txt",'r')
```

```
r=f.read()
```

```
print('Total number of character:',len(r))
```

```
l1=r.split()
```

```
print("Total number of words",len(l1))
```

```
l2=r.split("\n")
```

```
print(l2)
```

```
print('Total number of lines',len(l2)-1,'\n')
```

```
f.close()
```

```
#b
```

```
print('b)Calculating the frequency of each character in form of dictionary')
```

```
f=open("Text.txt",'r')
```

```
r=f.read()
```

```
freq={}
```

```
for i in r:
```

```
    if i in " \n":
```

```
        continue
```

```
    a=r.count(i)
```

```
    freq[i]=a
```

```
print('Frequency of each character is:',freq,'\n')
```

```
f.close()
#c
print('c)Printing words of file in reverse order')
f=open("Text.txt",'r')
rev=f.readlines()[::-1]
for i in rev:
    l=i.split()[::-1]
    for j in l:
        print(j,end=' ')
    print()
f.close()
#d
print('\nd)Copy even lines to File1.txt and odd lines to File2.txt')
f=open('Text.txt','r')
f1=open('File1.txt','w+')
f2=open('File2.txt','w+')
r=f.readlines()
c=1
for i in r:
    if c%2==0:
        f1.write(i)
    else:
        f2.write(i)
    c+=1
print('Copy Done')
f.seek(0)
f1.seek(0)
f2.seek(0)
print('Text.txt:')
print(f.read())
print('File1.txt:')
print(f1.read())
print('File2.txt:')
print(f2.read())
f.close()
f1.close()
f2.close()
```

Output:

```
File Name:Text.txt

File doesnot exist first create it
How many lines do you want to add:3
Enter line1:I am niket aggarwal.
Enter line2:I am in BSc(H) CS IN SSCBS
Enter line3:My roll no. is 25739
File created

a)Total no. of character,words and lines in the file
Total number of character: 69
Total number of words 16
['I am niket aggarwal.', 'I am in BSc(H) CS IN SSCBS', 'My roll no. is 25739', '']
Total number of lines 3

b)Calculating the frequency of each character in form of dictionary
Frequency of each character is: {'I': 3, 'a': 5, 'm': 2, 'n': 3, 'i': 3, 'k': 1, 'e': 1, 't': 1, 'g': 2, 'r': 2, 'w': 1, 'l': 3, '.': 2, 'B': 2, 'S': 5, 'C': 1, '(': 1, 'H': 1, ')': 1, ' ': 2, 'N': 1, 'M': 1, 'y': 1, 'o': 2, 's': 1, '2': 1, '5': 1, '7': 1, '3': 1, '9': 1}

c)Printing words of file in reverse order
25739 is no. roll My
SSCBS IN CS BSc(H) in am I
aggarwal. niket am I

d)Copy even lines to File1.txt and odd lines to File2.txt
```

```
d)Copy even lines to File1.txt and odd lines to File2.txt
Copy Done
Text.txt:
I am niket aggarwal.
I am in BSc(H) CS IN SSCBS
My roll no. is 25739

File1.txt:
I am in BSc(H) CS IN SSCBS

File2.txt:
I am niket aggarwal.
My roll no. is 25739
```

Q10- Write a function that prints a dictionary where keys are number between 1 and 5 and the values are the cubes of the keys.

Program:

```
def dict_cube():  
    d={}  
    for i in range(1,6):  
        d[i]=i**3  
    print("Dictionary:",d)  
print('A dictionary that habe key between 1 and 5 and value as cube of key')  
dict_cube()
```

Output:

```
===== RESTART: C:\Users\Shruti\Desktop\Niket Aggarwal\bsc ho  
A dictionary that habe key between 1 and 5 and value as cube of key  
Dictionary: {1: 1, 2: 8, 3: 27, 4: 64, 5: 125}  
|
```

Q11- Consider a tuple t1=(1,2,5,7,9,2,4,6,8,10). Write a program to perform following operations:

- a. Print half the values of the tuple in one line and half in the next line.
- b. Print another tuple whose values are even numbers in the given tuple.
- c. Concatenate a tuple t2=(11,13,15) with t1.
- d. Return maximum and minimum value from this tuple

Program:

```
print('Tuple Operations:',end='')
t1=(1,2,5,7,9,2,4,6,8,10)
print(t1)
print("a)Half in one line and other half in second line")
l=len(t1)//2
print("First half:",t1[:l])
print("second half:",t1[l:],'\n')
print("b)Another Tuple with even values")
t=()
for i in t1:
    if i%2==0:
        t=t+(i,)
print('New tuple:',t,'\n')
t2=(11,13,15)
print("c)Concatenate")
print("Original:",t1)
print("To add:",t2)
print("Cancatenated:",t1+t2,'\n')
print("d)Max and Min value of the Tuple")
print("Max:",max(t1))
print("Min:",min(t1))
```

Output:

```
Tuple Operations:(1, 2, 5, 7, 9, 2, 4, 6, 8, 10)
a)Half in one line and other half in second line
First half: (1, 2, 5, 7, 9)
second half: (2, 4, 6, 8, 10)

b)Another Tuple with even values
New tuple: (2, 2, 4, 6, 8, 10)

c)Concatenate
Original: (1, 2, 5, 7, 9, 2, 4, 6, 8, 10)
To add: (11, 13, 15)
Cancatenated: (1, 2, 5, 7, 9, 2, 4, 6, 8, 10, 11, 13, 15)

d)Max and Min value of the Tuple
Max: 10
Min: 1
```

Q12- Define a class Employee that stores information about employees in the company. The class should contain the following:

(i) Data members- count (to keep a record of all the objects being created for this class) and for every employee: an employee number, Name, Dept, Basic, DA and HRA.

(ii) Function members:

a. `__init__` method to initialize and/or update the members. Add statements to ensure that the program is terminated if any of Basic, DA and HRA is set to a negative value.

b. Function salary, that returns salary as the sum of Basic, DA and HRA.

c. `__del__` function to decrease the number of objects created for the class.

d. `__str__` function to display the details of an employee along with the salary of an employee in a proper format.

Program:

```
print("Class Employee is created to store information of Employee\n")
```

```
class Employee:
```

```
    count=0
```

```
    def __init__(self,empno,name,dept,basic,da,hra):
```

```
        if basic<0 or da<0 or hra<0:
```

```
            raise ValueError("Basic,DA and HRA must be non-negative values.")
```

```
        self.empno = empno
```

```
        self.name = name
```

```
        self.dept = dept
```

```
        self.basic = basic
```

```
        self.da = da
```

```
        self.hra = hra
```

```
        Employee.count+= 1
```

```
    def get_salary(self):
```

```
        return self.basic+self.da+self.hra
```

```
    def __del__(self):
```

```
        Employee.count-=1
```

```
    def __str__(self):
```

```
        return f"EmployeeNo:{self.empno}\nName:{self.name}\nDepartment:
```

```
{self.dept}\nBasic Pay:{self.basic}\nDA:{self.da}\nHRA:{self.hra}\nTotal Salary:
{self.get_salary()}"
```

```
try:
```

```
    E1=Employee(101,"Niket","AI Research",50000,10000,5000)
```

```
    E2=Employee(102,"Riya","Data Science",60000,12000,6000)
```

```
    print("Total Employee:",Employee.count)
```

```
    print(E1.__str__(),'\n')
```

```
    print(E2.__str__())
```

```
    del E1
```

```
    print("\nEmployee with id 101 id deleted')
```

```
print("Remaining Employees:",Employee.count)
print(E2.__str__(),'\n')
print(E1.__str__())
except ValueError as e:
    print("Error:", e)
except:
    print("Object not found")
```

Output:

```
Class Employee is created to store information of Employee
```

```
Total Employee: 2
EmployeeNo:101
Name:Niket
Department:AI Research
Basic Pay:50000
DA:10000
HRA:5000
Total Salary:65000

EmployeeNo:102
Name:Riya
Department:Data Science
Basic Pay:60000
DA:12000
HRA:6000
Total Salary:78000
```

```
Employee with id 101 id deleted
Remaining Employees: 1
EmployeeNo:102
Name:Riya
Department:Data Science
Basic Pay:60000
DA:12000
HRA:6000
Total Salary:78000

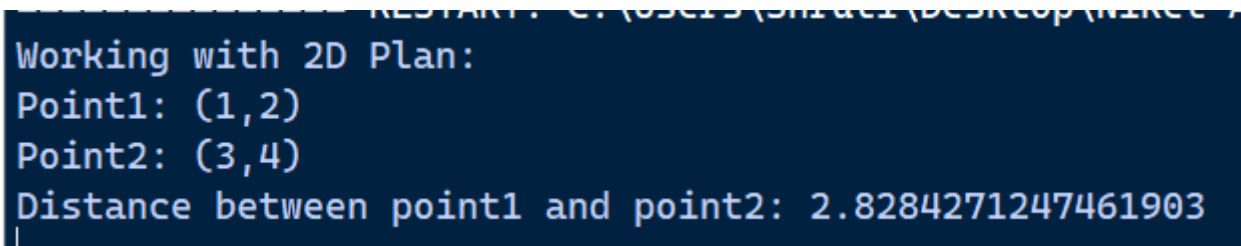
Object not found
```

Q13- Write a program to define a class "2DPoint" with coordinates x and y as attributes. Create relevant methods and print the objects. Also define a method distance to calculate the distance between any two point objects.

Program:

```
print("Working with 2D Plan:")
import math
class Point2D:
    def __init__(self,x,y):
        self.x=x
        self.y=y
    def get_coordinates(self):
        return f"({self.x},{self.y})"
    def dist_between(self,other):
        return math.sqrt((self.x-other.x)**2+(self.y-other.y)**2)
def main():
    P1=Point2D(1,2)
    P2=Point2D(3,4)
    print("Point1:",P1.get_coordinates())
    print("Point2:",P2.get_coordinates())
    print("Distance between point1 and point2:",P1.dist_between(P2))
if __name__=="__main__":
    main()
```

Output:



```
Working with 2D Plan:
Point1: (1,2)
Point2: (3,4)
Distance between point1 and point2: 2.8284271247461903
```

Q14- Inherit the above class to create a "3Dpoint" with additional attribute z. Override the method defined in "2DPoint" class , to calculate distance between two points of the "3DPoint" class.

Program:

#Above Class is already created

```
print("Working with 3D Plan:")
class Point3D(Point2D):
    def __init__(self,x,y,z):
        super().__init__(x,y)
        self.z=z
    def get_coordinates(self):
        return f"Point({self.x},{self.y},{self.z})"
    def distance(self,other):
        return math.sqrt((self.x-other.x)**2+(self.y-other.y)**2+(self.z-other.z)**2)
def main():
    P3=Point3D(1,2,3)
    P4=Point3D(4,6,9)
    print("Point3:",P3.get_coordinates())
    print("Point4:",P4.get_coordinates())
    print("Distance between point3 and point4:",P3.distance(P4))
if __name__=="__main__":
    main()
```

Output:

```
Working with 2D Plan:
Working with 3D Plan:
Point3: Point(1,2,3)
Point4: Point(4,6,9)
Distance between point3 and point4: 7.810249675906654
```

Q15- Write a program to accept a name from a user. Raise and handle appropriate exception(s) if the text entered by the user contains digits and/or special characters.

Program:

```
print("To check that the user enter name in correct format or not")
try:
    a=input("Enter Username:")
    if a=="" or a==" ":
        raise ValueError("Username field cannot be empty")
    for i in a:
        if i.isspace():
            pass
        elif i.isalpha()==False:
            raise ValueError("Given username is not correct it will not contain
number/special character")
    print("Entered username is correct that is:",a)
except ValueError as e:
    print("\nError:",e)
```

Output:

```
To check that the user enter name in correct format or not
Enter Username:niket@123

Error: Given username is not correct it will not contain number/special character
```

```
To check that the user enter name in correct format or not
Enter Username:Niket Aggarwal
Entered username is correct that is: Niket Aggarwal
```

```
To check that the user enter name in correct format or not
Enter Username:

Error: Username field cannot be empty
```